

LINEAR TRANSFORMATION

Dr. Juhi Kesarwani

Dr. Ashish Kumar Kesarwany

Mathematics Division, School of Advanced Sciences and Languages
VIT Bhopal University, Bhopal-Indore Highway, Kothrikalan, Sehore, Madhya Pradesh, 466114, India

October 22, 2024



VIT[®]
BHOPAL
www.vitbhopal.ac.in

Table of Contents

- 1 Linear Transformation
- 2 Kernel and Range
- 3 The Matrix of a Linear Transformation
- 4 Vector space of linear transformations
- 5 Change of basis
- 6 Similarity
- 7 Application of Linear transformations in Computer graphics

Linear Transformation

✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$.

Linear Transformation

✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$.

Linear Transformation

✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$. Thus, we have a map (function) $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(v) = Av.$$

Linear Transformation

- ✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$. Thus, we have a map (function) $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(v) = Av.$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ given by

$$F(p(x)) = \frac{d}{dx} p(x).$$

Linear Transformation

- ✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$. Thus, we have a map (function) $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(v) = Av.$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ given by

$$F(p(x)) = \frac{d}{dx} p(x).$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}$ given by

$$F(p(x)) = p(3).$$

Linear Transformation

- ✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$. Thus, we have a map (function) $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(v) = Av.$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ given by

$$F(p(x)) = \frac{d}{dx} p(x).$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}$ given by

$$F(p(x)) = p(3).$$

What is common in all of these?

Linear Transformation

- ✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$. Thus, we have a map (function) $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(v) = Av.$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ given by

$$F(p(x)) = \frac{d}{dx} p(x).$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}$ given by

$$F(p(x)) = p(3).$$

What is common in all of these? Well, they are maps (functions) with domains and co-domains as vector spaces over $\mathbb{R}$.

Linear Transformation

- ✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$. Thus, we have a map (function) $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(v) = Av.$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ given by

$$F(p(x)) = \frac{d}{dx} p(x).$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}$ given by

$$F(p(x)) = p(3).$$

What is common in all of these? Well, they are maps (functions) with domains and co-domains as vector spaces over $\mathbb{R}$. What else?

Linear Transformation

- ✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$. Thus, we have a map (function) $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(v) = Av.$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ given by

$$F(p(x)) = \frac{d}{dx} p(x).$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}$ given by

$$F(p(x)) = p(3).$$

What is common in all of these? Well, they are maps (functions) with domains and co-domains as vector spaces over $\mathbb{R}$. What else? We have

$$F(u + v) = F(u) + F(v),$$

Linear Transformation

- ✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$. Thus, we have a map (function) $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(v) = Av.$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ given by

$$F(p(x)) = \frac{d}{dx} p(x).$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}$ given by

$$F(p(x)) = p(3).$$

What is common in all of these? Well, they are maps (functions) with domains and co-domains as vector spaces over $\mathbb{R}$. What else? We have

$$F(u + v) = F(u) + F(v), \quad F(\alpha v) = \alpha F(v),$$

or, equivalently,

Linear Transformation

- ✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$. Thus, we have a map (function) $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(v) = Av.$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ given by

$$F(p(x)) = \frac{d}{dx} p(x).$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}$ given by

$$F(p(x)) = p(3).$$

What is common in all of these? Well, they are maps (functions) with domains and co-domains as vector spaces over $\mathbb{R}$. What else? We have

$$F(u + v) = F(u) + F(v), \quad F(\alpha v) = \alpha F(v),$$

or, equivalently, $F(\alpha u + v) = \alpha F(u) + F(v)$.

Linear Transformation

- ✿ Suppose $A \in \mathcal{M}_{m \times n}$. Take $v \in \mathbb{R}^n$. Then $Av \in \mathbb{R}^m$. Thus, we have a map (function) $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$F(v) = Av.$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ given by

$$F(p(x)) = \frac{d}{dx} p(x).$$

- ✿ Take $F : \mathbb{R}[x] \rightarrow \mathbb{R}$ given by

$$F(p(x)) = p(3).$$

What is common in all of these? Well, they are maps (functions) with domains and co-domains as vector spaces over $\mathbb{R}$. What else? We have

$$F(u + v) = F(u) + F(v), \quad F(\alpha v) = \alpha F(v),$$

or, equivalently, $F(\alpha u + v) = \alpha F(u) + F(v)$. Such functions are called **linear transformations (LT)**.

Linear Transformation

Definition(Linear Transformation)

A **linear transformation** from a vector space $V(\mathbb{F})$ into a vector space $W(\mathbb{F})$ is a mapping $T : V \rightarrow W$ such that for all $u, v \in V$ and for all $a \in \mathbb{F}$

$$T(au + v) = aT(u) + T(v).$$

Equivalently, $T(u + v) = T(u) + T(v)$ and $T(av) = aT(v)$ for all $u, v \in V, a \in \mathbb{F}$.

Example

Let A be an $m \times n$ matrix. Define $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that $T(x) = Ax$ for all $x \in \mathbb{R}^n$. Then T is a linear transformation from $\mathbb{R}^n$ into $\mathbb{R}^m$.

Example

The map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x \\ x + y \end{bmatrix}$ for all $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$, is a linear transformation.

Example

Let V and W be two vector spaces over $\mathbb{F}$. The map $T_0 : V \rightarrow W$, defined by $T_0(v) = 0$ for all $v \in V$, is a linear transformation. The map T_0 is called the **zero transformation**.

Example

Let V be a vector space over $\mathbb{F}$. The map $I : V \rightarrow V$, defined by $I(v) = v$ for all $v \in V$, is a linear transformation. The map I is called the **identity transformation**.

Example

The map $T : \mathbb{R} \rightarrow \mathbb{R}$, defined by $T(x) = x + 1$ for all $x \in \mathbb{R}$, is **not** a linear transformation.

Result

Let $T : V \rightarrow W$ be a linear transformation. Then

- ① $T(0) = 0$;
- ② $T(-v) = -T(v)$ for all $v \in V$; and
- ③ $T(u - v) = T(u) - T(v)$ for all $u, v \in V$.

Result

Let $\{v_1, \dots, v_n\}$ be a basis of V ($\dim(V) = n$). Let $u_1, \dots, u_n$ be arbitrarily chosen in W . Then there is a unique linear transformation $T : V \rightarrow W$ such that $T(v_i) = u_i$.

Example

Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}_2[x]$ is a linear transformation such that $T \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = 2 - 3x + x^2$ and $T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = 1 - x^2$. Find $T \left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} \right)$ and $T \left(\begin{bmatrix} a \\ b \end{bmatrix} \right)$.

Example

Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}_2[x]$ is a linear transformation such that $T \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = 2 - 3x + x^2$ and $T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = 1 - x^2$. Find $T \left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} \right)$ and $T \left(\begin{bmatrix} a \\ b \end{bmatrix} \right)$.

Solution

$$\begin{aligned} \begin{bmatrix} a \\ b \end{bmatrix} &= a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} \implies T \left(\begin{bmatrix} a \\ b \end{bmatrix} \right) = T \left(a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \\ &= aT \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) + bT \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \\ &= a(2 - 3x + x^2) + b(1 - x^2) \\ &= (a - b)x^2 - 3ax + 2a + b \end{aligned}$$

Thus, $T \left(\begin{bmatrix} a \\ b \end{bmatrix} \right) = (a - b)x^2 - 3ax + 2a + b$ and hence $T \left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} \right) = -x^2 - 6x + 7$.

Exercise 1:

Examine whether the following maps $T : V \rightarrow W$ are linear transformations.

$$(a) \quad V = W = \mathbb{R}^3, \quad T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} 3x + y \\ z \\ |x| \end{bmatrix}.$$

$$(b) \quad V = W = \mathbb{R}^3, \quad T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} x - y + 5 \\ z^2 \\ xyz \end{bmatrix}.$$

$$(c) \quad V = \mathbb{R}^3, \quad W = \mathbb{R}^2, \quad T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} x - y + z \\ 2z - 3y + x \end{bmatrix}.$$

$$(d) \quad V = W = \mathbb{R}^2, \quad T \text{ is the projection onto } Y\text{-axis}.$$

Exercise 2:

Suppose $b, c \in \mathbb{R}$. Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} 2x - 4y + 3z + b \\ 6x + cxyz \end{bmatrix}$

Show that T is linear transformation if and only if $b = c = 0$.

Exercise 3:

Determine whether the function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x+y \\ x+1 \\ 3y \end{bmatrix}$$

is a linear transformation.

Exercise 4:

Let T be a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$ such that

$$T \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ and } T \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then find $T \left(\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right)$.

Exercise 5:

Suppose that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation satisfying

$$T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \text{ and } T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Find a general formula for $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right)$.

Exercise 6:

Let $\{\mathbf{v}_1, \mathbf{v}_2\}$ be a basis of the vector space $\mathbb{R}^2$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

The action of a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ on the basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ is given by

$$T(\mathbf{v}_1) = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \text{ and } T(\mathbf{v}_2) = \begin{bmatrix} 0 \\ 8 \\ 10 \end{bmatrix}.$$

Find the formula of $T(\mathbf{x})$, where

Exercise 7:

Let T be the linear transformation from the 3-dimensional vector space $\mathbb{R}^3$ to $\mathbb{R}^3$ itself satisfying the following relations.

$$T \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad T \left(\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \quad T \left(\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Then for any vector $\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3$, find the formula for $T(\mathbf{x})$.

Exercise 8:

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation such that

$$T(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, T(\mathbf{e}_2) = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, T(\mathbf{e}_3) = \begin{bmatrix} 3 \\ 6 \end{bmatrix},$$

where $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are the standard unit basis vectors of $\mathbb{R}^3$.

For any vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3$, find a formula for $T(\mathbf{x})$.

Exercise 9:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}_2[x]$ be a linear map for which $T\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = 1 + 2x + x^2$ and $T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = 3 + 4x^2$. Find $T\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right)$ and $T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right)$.

Exercise 10:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation. Let $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ be 2-dimensional vectors. Suppose that

$$T(\mathbf{u}) = T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} -3 \\ 5 \end{bmatrix},$$

$$T(\mathbf{v}) = T\left(\begin{bmatrix} 3 \\ 5 \end{bmatrix}\right) = \begin{bmatrix} 7 \\ 1 \end{bmatrix}.$$

Let $\mathbf{w} = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$. Find the formula for $T(\mathbf{w})$ in terms of x and y .

Exercise 11:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation such that $T(\mathbf{e}_1) = \mathbf{u}_1$ and $T(\mathbf{e}_2) = \mathbf{u}_2$, where $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are unit vectors of $\mathbb{R}^2$ and

$$\mathbf{u}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}. \text{ Then find } T\left(\begin{bmatrix} 3 \\ -2 \end{bmatrix}\right).$$

Exercise 12:

If $L : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation such that

$$L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad L\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix}.$$

then

(a) find $L\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right)$, and

(b) find the formula for $L\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$.

If you think you can solve (b), then skip (a) and solve (b) first and use the result of (b) to answer (a).

Exercise 13:

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}_2[x]$ be a linear map for which $T\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = 1 + 2x + x^2$ and $T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = 3 + 4x^2$. Find $T\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right)$ and $T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right)$.

Composition of Linear Transformation

Composition of Linear Transformation

Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be two linear transformations. The **composition** of S with T is the mapping $S \circ T : U \rightarrow W$ defined by

$$(S \circ T)(u) = S(T(u)) \text{ for all } u \in U.$$

Result

Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be two linear transformations. Then the composition $S \circ T : U \rightarrow W$ is also a **linear transformation**.

Example

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} x + 2z \\ 3x + y \\ y - z \end{bmatrix}$.

Find T^2 .

Inverse of a function

Inverse of a function

A function $f : X \rightarrow Y$ is said to be **invertible** if there is another function $g : Y \rightarrow X$ such that

$$g \circ f = I_X \text{ and } f \circ g = I_Y.$$

- ✿ If f is invertible, the the function g satisfying $g \circ f = I_X$, $f \circ g = I_Y$ is called **inverse of f** .
- ✿ Inverse of a function, if it exists, is **unique**.
- ✿ The symbol f^{-1} is used to denote **the inverse** of f .
- ✿ **Inverse** of a linear transformation is **linear**.

Example

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x - y \\ -3x + 4y \end{bmatrix}$$

and

$$S \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} 4x + y \\ 3x + y \end{bmatrix}$$

for all $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$. Then S is the inverse of T .

Kernel and Range

Definition(Kernel and Range)

Let $T : V \rightarrow W$ be a linear transformation. Then the **kernel** of T (null space of T), denoted $\ker(T)$, and the **range** of T , denoted $\text{range}(T)$, are defined as

$$\ker(T) = \{v \in V : T(v) = 0\}, \text{ and}$$

$$\text{range}(T) = \{T(v) : v \in V\}.$$

Result

Let V and W be vector spaces over $\mathbb{F}$ and let $T : V \rightarrow W$ be a linear transformation. Then $\ker(T)$ is a **subspace** of V and $\text{range}(T)$ is a **subspace** of W .

Definition(Rank and Nullity of a linear transformation)

Let $T : V \rightarrow W$ be a linear transformation. We define

- ✿ $\text{rank}(T) = \text{dimension of } \text{range}(T)$; and
- ✿ $\text{nullity}(T) = \text{dimension of } \ker(T)$.

Example

Let A be an $m \times n$ matrix. Define $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that $T(x) = Ax$ for all $x \in \mathbb{R}^n$. Then $\ker(T) = \mathcal{N}(A)$ and $\text{range}(T) = \mathcal{C}(A)$.

Example

Let $D : \mathbb{R}_3[x] \rightarrow \mathbb{R}_2[x]$ be defined by $D(p(x)) = \frac{d}{dx}p(x)$. Then $\text{rank}(D) = 3$ and $\text{nullity}(D) = 1$.

Result (The Rank-Nullity Theorem)

Let $T : V \rightarrow W$ be a linear transformation from a **finite dimensional** vector space V into a vector space W . Then

$$\text{rank}(T) + \text{nullity}(T) = \dim(V).$$

Exercise 14:

For each of the maps given below, show that T is a linear transformation. Describe $\ker(T)$ and $\text{range}(T)$ of the following. Hence find the rank or the nullity of the following

(a) $T : M_2(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $T(A) = \text{tr}(A)$.

(b) $T : \mathbb{R}_3[x] \rightarrow \mathbb{R}^3$, $T(a + bx + cx^2 + dx^3) = \begin{bmatrix} a \\ b - a \\ c - a + b \end{bmatrix}$.

(c) $T : M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ defined by $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$.

(d) $T : M_2(\mathbb{R}) \rightarrow \mathbb{R}^2$ defined by $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a - b \\ c - d \end{bmatrix}$.

(e) $T : \mathbb{R}_2[x] \rightarrow \mathbb{R}^2$ defined by $T(p(x)) = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix}$.

Exercise 15:

For an integer $n > 0$, let $\mathbb{R}_n[x]$ be the vector space of polynomials of degree at most n . The set $B = \{1, x, x^2, \dots, x^n\}$ is a basis of $\mathbb{R}_n[x]$, called the standard basis.

Let $T : \mathbb{R}_n[x] \rightarrow \mathbb{R}_{n+1}[x]$ be the map defined by

$$T(p(x)) = xp(x).$$

Prove that T is a linear transformation, and find its range and nullspace.

Exercise 16:

Let V denote the vector space of 2×2 matrices, and W the vector space of 3×2 matrices. Define the linear transformation $T : V \rightarrow W$ by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a+b & 2d \\ 2b-d & -3c \\ 2b-c & -3a \end{bmatrix}.$$

Find a basis for the range(T).

Exercise 17:

Let $M_2(\mathbb{R})$ be the vector space of 2×2 matrices with real entries, and $\mathbb{R}_3[x]$ the vector space of real polynomials of degree 3 or less. Define the linear transformation $T : M_2(\mathbb{R}) \rightarrow \mathbb{R}_3[x]$ by

$$T \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = 2a + (b - d)x - (a + c)x^2 + (a + b - c - d)x^3.$$

Find the rank and nullity of T .

Solution

ker(T):

$$\begin{aligned}\ker(T) &= \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_2(\mathbb{R}) : T \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = 0 \right\} \\ &= \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_2(\mathbb{R}) : 2a + (b - d)x - (a + c)x^2 + (a + b - c - d)x^3 = 0 \right\} \\ &= \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_2(\mathbb{R}) : a = c = 0 \text{ \& } b = d \right\} \\ &= \left\{ \begin{bmatrix} 0 & b \\ 0 & b \end{bmatrix} : b \in \mathbb{R} \right\} = \left\{ b \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} : b \in \mathbb{R} \right\} = \text{span} \left(\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \right)\end{aligned}$$

Clearly, $\left\{ \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \right\}$ is a basis of $\ker(T)$. Therefore $\text{nullity}(T) = \dim(\ker(T)) = 1$.

range(T):

$$\begin{aligned}\text{range}(T) &= \left\{ T \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) : \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_2(\mathbb{R}) \right\} \\&= \{ 2a + (b-d)x - (a+c)x^2 + (a+b-c-d)x^3 : a, b, c, d \in \mathbb{R} \} \\&= \{ a(2-x^2+x^3) + c(-x^2-x^3) + (b-d)(x+x^3) : a, b, c, d \in \mathbb{R} \} \\&= \{ a(2-x^2+x^3) + c(-x^2-x^3) + \beta(x+x^3) : a, c, \beta \in \mathbb{R} \} \\&= \text{span}(2-x^2+x^3, -x^2-x^3, x+x^3)\end{aligned}$$

Clearly, $\{2-x^2+x^3, -x^2-x^3, x+x^3\}$ is a basis of $\text{range}(T)$. Therefore $\text{rank}(T) = \dim(\text{range}(T)) = 3$.

Exercise 18:

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation such that

$$T(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, T(\mathbf{e}_2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, T(\mathbf{e}_3) = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

where $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ are the standard basis of $\mathbb{R}^3$. Then find the rank and the nullity of T .

Exercise 19:

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} x + 2z \\ 3x + y \\ y - z \end{bmatrix}$.

Find the rank and nullity of T^2 .

Solution

First of all, we need to find the formula of T^2 .

$$T^2 \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = T \left(T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) \right) = T \left(\begin{bmatrix} x + 2z \\ 3x + y \\ y - z \end{bmatrix} \right) = \begin{bmatrix} x + 2y \\ 6x + y + 6z \\ 3x + z \end{bmatrix}$$

Therefore

$$\begin{aligned} \ker(T^2) &= \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : T^2 \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\} \\ &= \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : \begin{bmatrix} x + 2y \\ 6x + y + 6z \\ 3x + z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\} \\ &= \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x + 2y = 0, 6x + y + 6z = 0, \& 3x + z = 0 \right\} \\ &= \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x = 0, y = 0, \& z = 0 \right\} = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\} \end{aligned}$$

Thus,

$$\text{nullity}(T^2) = \dim(\ker(T^2)) = 0.$$

Now, from rank-nullity theorem

$$\begin{aligned}\text{rank}(T^2) + \text{nullity}(T^2) &= \dim(\mathbb{R}^3) = 3 \\ \implies \text{rank}(T^2) &= 3 - \text{nullity}(T^2) = 3 - 0 = 3\end{aligned}$$

Therefore $\text{rank}(T^2) = 2$ and $\text{nullity}(T^2) = 0$.

One-one and Onto Linear Transformation

Definition (One-one Linear Transformation)

Let $T : V \rightarrow W$ be a linear transformation. Then

✿ T is called **one-one** if T maps distinct vectors in V into distinct vectors in W .

OR

✿ For all $u, v \in V$, if $u \neq v$ implies that $T(u) \neq T(v)$, then T is one-one.

OR

✿ For all $u, v \in V$, if $T(u) = T(v)$ implies that $u = v$, then T is one-one.

Result

A linear transformation $T : V \rightarrow W$ is **one-one** iff $\ker(T) = \{0\}$.

Definition(Onto Linear Transformation)

Let $T : V \rightarrow W$ be a linear transformation. Then

- ✿ For all $w \in W$, if there is at least one $v \in V$ such that $T(v) = w$, then T is onto.

OR

- ✿ T is called onto if $\text{range}(T) = W$.

Example

$T : \mathbb{R} \rightarrow \mathbb{R}^2$ defined by $T(x) = \begin{bmatrix} x \\ 0 \end{bmatrix}$ for $x \in \mathbb{R}$ is **one-one** but **not onto**.

Example

$T : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = x$, for $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$ is **onto** but **not one-one**.

Example

$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} -x \\ -y \end{bmatrix}$, for $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$ is **one-one** and **onto**.

Result

Let $\dim(V) = \dim(W)$. Then a linear transformation $T : V \rightarrow W$ is **one-one** iff T is **onto**.

Result

Let $\dim(V) = \dim(W)$. Then a linear transformation $T : V \rightarrow W$ is **one-one** iff T is **onto**.

Example

Let $T : V \rightarrow W$ be a linear transformation and $v_1, \dots, v_k \in V$ such that $T(v_1), \dots, T(v_k)$ are **linearly independent**. Can $v_1, \dots, v_k$ be **linearly dependent**? Justify?

Exercise 20:

Let $T : V \rightarrow W$ be a one-one linear transformation. If $S = \{v_1, \dots, v_k\}$ is a linearly independent set in V then $T(S) = \{T(v_1), \dots, T(v_k)\}$ is a linearly independent set in W .

Definition(Isomorphism)

- ❁ A linear transformation $T : V \rightarrow W$ is called an **isomorphism** if it is **one-one** and **onto**.
- ❁ If $T : V \rightarrow W$ is an isomorphism then we say that V and W are isomorphic, and we write $V \sim W$.

Example

The vector spaces $\mathbb{R}_3$ and $\mathbb{R}_2[x]$ are **isomorphic**.

Result

Let $V(\mathbb{F})$ and $W(\mathbb{F})$ be two finite dimensional vector spaces. Then V is **isomorphic** to W iff $\dim(V) = \dim(W)$.

Example

The vector spaces $\mathbb{R}^n$ and $\mathbb{R}_n[x]$ are **not** isomorphic.

The Matrix of a Linear Transformation

Let V and W be two vector spaces with ordered bases $\mathcal{B}$ and $\mathcal{C}$ respectively, where $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ and $\mathcal{C} = \{u_1, u_2, \dots, u_m\}$. Let $T : V \rightarrow W$ be a linear transformation, then $T(v_1), \dots, T(v_n) \in W$ and $\mathcal{C}$ is a basis of W , So let

$$T(v_1) = a_{11}u_1 + a_{21}u_2 + \dots + a_{m1}u_m$$

$$T(v_2) = a_{12}u_1 + a_{22}u_2 + \dots + a_{m2}u_m$$

$$\vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots$$

$$T(v_n) = a_{1n}u_1 + a_{2n}u_2 + \dots + a_{mn}u_m$$

then the $m \times n$ matrix A defined by

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

is called the **matrix of T with respect to the ordered bases $\mathcal{B}$ and $\mathcal{C}$** and is denoted by $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$. If $\mathcal{B} = \mathcal{C}$, then $[T]_{\mathcal{B} \leftarrow \mathcal{B}}$ is simply written as $[T]_{\mathcal{B}}$.

Remark

Suppose we know $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$ w.r.t. given ordered bases $\mathcal{B}$ and $\mathcal{C}$. Then we know T in the following sense:

$$[T(v)]_{\mathcal{C}} = [T]_{\mathcal{C} \leftarrow \mathcal{B}}[v]_{\mathcal{B}}$$

Example

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} x - 2y \\ x + y - 3z \end{bmatrix}$ for $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3$. Let $\mathcal{B} = \{e_1, e_2, e_3\}$ and $\mathcal{C} = \{e_2, e_1\}$ be ordered bases for $\mathbb{R}^3$ and $\mathbb{R}^2$, respectively. Find $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$ and verify $[T(v)]_{\mathcal{C}} = [T]_{\mathcal{C} \leftarrow \mathcal{B}}[v]_{\mathcal{B}}$ for $v = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$.

Example

Consider $D : \mathbb{R}_3[x] \rightarrow \mathbb{R}_3[x]$ defined by $D(p(x)) = p'(x)$. Find the matrix $[D]_{\mathcal{B}}$ with respect to the ordered basis $\mathcal{B} = \{1, x, x^2, x^3\}$ of $\mathbb{R}_3[x]$. Also verify that $[D]_{\mathcal{B}}[p(x)]_{\mathcal{B}} = [D(p(x))]_{\mathcal{B}}$.

Example

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the transformation that rotates each point in $\mathbb{R}^2$ about the origin through an angle ϕ with counter-clockwise rotation for a positive angle. Find the matrix of linear transformation with respect to standard basis.

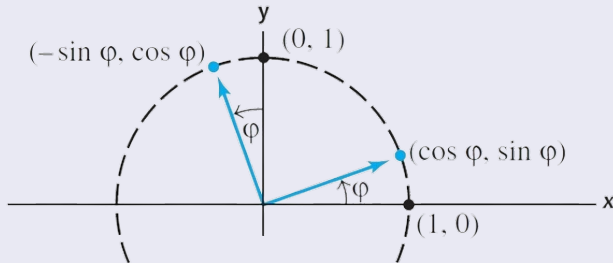


Figure 1: A rotation transformation

From above figure it is clear that

$$T \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} \cos \phi \\ \sin \phi \end{bmatrix} = \cos \phi \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \sin \phi \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

$$T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} \cos \left(\frac{\pi}{2} + \phi \right) \\ \sin \left(\frac{\pi}{2} + \phi \right) \end{bmatrix} = \begin{bmatrix} -\sin \phi \\ \cos \phi \end{bmatrix} = -\sin \phi \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \cos \phi \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore the matrix of linear transformation with respect to usual basis is given by

$$\begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$

Result

Let U, V and W be three vector spaces with ordered bases $\mathcal{B}, \mathcal{C}$ and $\mathcal{D}$, respectively. Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear transformations. Then

$$[S \circ T]_{\mathcal{D} \leftarrow \mathcal{B}} = [S]_{\mathcal{D} \leftarrow \mathcal{C}} [T]_{\mathcal{C} \leftarrow \mathcal{B}}.$$

Result

Let $T : V \rightarrow W$ be a linear transformation between two n -dimensional vector spaces V and W with ordered bases $\mathcal{B}$ and $\mathcal{C}$, respectively. Then T is invertible if and only if the matrix $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$ is invertible. In this case,

$$([T]_{\mathcal{C} \leftarrow \mathcal{B}})^{-1} = [T^{-1}]_{\mathcal{B} \leftarrow \mathcal{C}}.$$

Example

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}_1[x]$ be defined by

$$T \left(\begin{bmatrix} a \\ b \end{bmatrix} \right) = a + (a + b)x$$

for $\begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2$. Show that T is invertible, and hence find T^{-1} .

Exercise 21:

In each of the following, find the matrix of T with respect to the standard bases. Is the matrix invertible? If so, find the inverse matrix and hence the inverse linear map of T .

$$(i) \quad T: \mathbb{R}^2 \rightarrow \mathbb{R}_1[x], \quad T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = a + (a - b)x.$$

$$(ii) \quad D: \text{Span}(e^x, xe^x, x^2e^x) \rightarrow \mathcal{F} \text{ where } D \text{ is the differential operator and } \mathcal{F} \text{ is the space of all real valued functions from } \mathbb{R} \text{ to } \mathbb{R}.$$

Exercise 22:

Find the matrix $[S]$, $[T]$ and $[S \circ T]$ with respect to the standard bases and then verify that $[S \circ T] = [S][T]$, where

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}_2[x], \quad T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = a + (a - b)x + bx^2$$

$$S: \mathbb{R}_2[x] \rightarrow \mathbb{R}_1[x], \quad S(a + bx + cx^2) = (3a + 2b + c) + (a + b)x.$$

Exercise 23:

Define the map $T : \mathbb{R}_2[x] \rightarrow \mathbb{R}_4[x]$ by

$$T(p(x)) = p(x^2).$$

Determine if T is a linear transformation. If it is, find the matrix representation for T relative to the basis $\mathcal{B} = \{1, x, x^2\}$ of $\mathbb{R}_2[x]$ and $\mathcal{C} = \{1, x, x^2, x^3, x^4\}$ of $\mathbb{R}_4[x]$.

Exercise 24:

Let $T : \mathbb{R}_4[x] \rightarrow \mathbb{R}_4[x]$ be the map defined by, for $p(x) \in \mathbb{R}_4[x]$,

$$T(p(x)) = p(x) - x - 1.$$

Determine if $T(x)$ is a linear transformation. If it is, find the matrix representation of T relative to the standard basis of $\mathbb{R}_4[x]$.

Exercise 25:

Let $T : \mathbb{R}_3[x] \rightarrow \mathbb{R}_5[x]$ be the map defined by, for $p(x) \in \mathbb{R}_3[x]$,

$$T(p(x)) = (x^2 - 2)p(x).$$

Determine if T is a linear transformation. If it is, find the matrix representation of T relative to the standard basis of $\mathbb{R}_3[x]$ and $\mathbb{R}_5[x]$.

Exercise 26:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation such that

$$T\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ and } T\left(\begin{bmatrix} 4 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ -5 \\ 1 \end{bmatrix}.$$

- (a) Find the matrix representation of T (with respect to the standard basis for $\mathbb{R}^2$).
- (b) Determine the rank and nullity of T .

Exercise 27:

Let $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ be a basis for the vector space $\mathbb{R}^2$, and let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that

$$T(\mathbf{v}_1) = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \text{ and } T(\mathbf{v}_2) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

If $\mathbf{e}_1 = \mathbf{v}_1 + 2\mathbf{v}_2$ and $\mathbf{e}_2 = 2\mathbf{v}_1 - \mathbf{v}_2$, where $\mathbf{e}_1, \mathbf{e}_2$ are the standard unit vectors in $\mathbb{R}^2$, then find the matrix of T with respect to the basis $\{\mathbf{e}_1, \mathbf{e}_2\}$.

Exercise 28:

Let $\mathbb{R}_2[x]$ be the vector space over $\mathbb{R}$ consisting of all polynomials with real coefficients of degree 2 or less. Let $B = \{1, x, x^2\}$ be a basis of the vector space $\mathbb{R}_2[x]$. For each linear transformation $T : \mathbb{R}_2[x] \rightarrow \mathbb{R}_2[x]$ defined below, find the matrix representation of T with respect to the basis B . For $\mathbb{R}_2[x]$, define T as follows.

(a)

$$T(f(x)) = \frac{d^2}{dx^2} f(x) - 3 \frac{d}{dx} f(x)$$

(b)

$$T(f(x)) = \int_{-1}^1 (t - x)^2 f(t) dt$$

(c)

$$T(f(x)) = e^x \frac{d}{dx} (e^{-x} f(x))$$

Exercise 29:

Let $\mathbb{Q}$ denote the set of rational numbers (i.e., fractions of integers). Let V denote the set of the form $x + y\sqrt{2}$ where $x, y \in \mathbb{Q}$. You may take for granted that the set V is a vector space over the field $\mathbb{Q}$.

- (a) Show that $B = \{1, \sqrt{2}\}$ is a basis for the vector space V over $\mathbb{Q}$.
(b) Let $\alpha = a + b\sqrt{2} \in V$, and let $T_\alpha : V \rightarrow V$ be the map defined by

$$T_\alpha(x + y\sqrt{2}) := (ax + 2by) + (ay + bx)\sqrt{2} \in V$$

for any $x + y\sqrt{2} \in V$. Show that T_α is a linear transformation.

- (c) Let $\begin{bmatrix} x \\ y \end{bmatrix}_B = x + y\sqrt{2}$. Find the matrix T_B such that

$$T_\alpha(x + y\sqrt{2}) = \left(T_B \begin{bmatrix} x \\ y \end{bmatrix}_B \right),$$

and compute $\det T_B$.

Exercise 30:

Suppose that a real matrix A maps each of the following vectors

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

into the vectors

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix}, \mathbf{y}_3 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix},$$

respectively. That is, $A\mathbf{x}_i = \mathbf{y}_i$ for $i = 1, 2, 3$. Find the matrix A .

Exercise 31:

For the vector spaces $\mathbb{R}_1[x]$ and $\mathbb{R}^2$, choose the basis $\mathcal{B} = \{1, x\}$ for $\mathbb{R}_1[x]$ and $\mathcal{C} = \{e_1, e_2\}$ for $\mathbb{R}^2$. Let $T : \mathbb{R}_1[x] \rightarrow \mathbb{R}^2$ be the linear transformation defined by

$$T(a + bx) = \begin{bmatrix} a \\ a + b \end{bmatrix}$$

(i) Show that T is invertible.

(ii) Find $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$ and $[T^{-1}]_{\mathcal{B} \leftarrow \mathcal{C}}$

Exercise 32:

Let $\mathcal{B}$ be the standard basis of $\mathbb{R}^3$, and $S, T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be two linear transformations given by

$$\begin{aligned} S(e_1) &= \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, & S(e_2) &= \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, & S(e_3) &= \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \\ T(e_1) &= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, & T(e_2) &= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, & T(e_3) &= \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \end{aligned}$$

Compute $[S + T]_{\mathcal{B}}$, $[2T - S]_{\mathcal{B}}$, $[T \circ S]_{\mathcal{B}}$.

Exercise 33:

Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be the linear transformation defined by

$$T \left(\begin{bmatrix} x \\ y \\ z \\ u \end{bmatrix} \right) = \begin{bmatrix} x + 2y \\ x - 3z + u \\ 2y + 3z + 4u \end{bmatrix}. \text{ Let } \mathcal{B} \text{ and } \mathcal{C} \text{ be the standard bases for } \mathbb{R}^4 \text{ and }$$

$\mathbb{R}^3$, respectively. Find $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$.

Exercise 34:

Let $T : \mathbb{R}_2[x] \rightarrow \mathbb{R}_2[x]$ be the linear transformation defined by

$$T(p(x)) = (3 + x) \frac{d}{dx} p(x) + 2p(x),$$

and $S : \mathbb{R}_2[x] \rightarrow \mathbb{R}^3$ defined by

$$S(a + bx + cx^2) = \begin{bmatrix} a - b \\ a + b \\ c \end{bmatrix}.$$

For a basis $\mathcal{B} = \{1, x, x^2\}$ for $\mathbb{R}_2[x]$ and the standard basis $\mathcal{C} = \{e_1, e_2, e_3\}$ for $\mathbb{R}^3$, compute $[T]_{\mathcal{C}}$, $[S]_{\mathcal{C} \leftarrow \mathcal{B}}$ and $[S \circ T]_{\mathcal{C} \leftarrow \mathcal{B}}$.

Exercise 35:

Find the matrix representation of each of the following linear transformations T of $\mathbb{R}^3$ with respect to the standard basis $\mathcal{B} = \{e_1, e_2, e_3\}$, and $\mathcal{C} = \{e_3, e_2, e_1\}$:

$$(a) \quad T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} 2x - 3y + 4z \\ 5x - y + 2z \\ 4x + 7y \end{bmatrix}, \quad (b) \quad T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} 2y + z \\ x - 4y \\ 3x \end{bmatrix}.$$

Vector space of linear transformations

Note

- ✿ The set of all linear transformations from a vector space V to a vector space W is denoted by $\mathcal{L}(V, W)$.
- ✿ The set of linear maps from V to V is denoted by $\mathcal{L}(V)$. In other words, $\mathcal{L}(V) = \mathcal{L}(V, V)$.

Definition(Addition and Scalar multiplication on $\mathcal{L}(V, W)$)

Suppose $S, T \in \mathcal{L}(V, W)$ and $\alpha \in \mathbb{F}$. The sum $S + T$ and the product αT are the linear transformations from V to W defined by

$$(S + T)(v) = S(v) + T(v) \quad \text{and} \quad (\alpha T)(v) = \alpha(T(v))$$

for all $v \in V$.

Result

With the operations of addition and scalar multiplication defined as

$$(S + T)(v) = S(v) + T(v) \quad \text{and} \quad (\alpha T)(v) = \alpha(T(v))$$

$\forall S, T \in \mathcal{L}(V, W)$, $\forall v \in V$ and $\forall \alpha \in \mathbb{F}$, $\mathcal{L}(V, W)$ becomes a vector space over $\mathbb{F}$.

Exercise 36:

Prove that $\mathcal{L}(V, W)$ is a vector space.

Change of basis

Definition(Transition matrix or change of basis matrix)

Let $\mathcal{B} = \{u_1, u_2, \dots, u_n\}$ and $\mathcal{C} = \{v_1, v_2, \dots, v_n\}$ be ordered bases for a vector space V . The $n \times n$ matrix $[I]_{\mathcal{C} \leftarrow \mathcal{B}}$ is called the **Transition matrix from $\mathcal{B}$ to $\mathcal{C}$** or **change of basis matrix from $\mathcal{B}$ to $\mathcal{C}$** and the $n \times n$ matrix $[I]_{\mathcal{B} \leftarrow \mathcal{C}}$ is called the **Transition matrix from $\mathcal{C}$ to $\mathcal{B}$** or **change of basis matrix from $\mathcal{C}$ to $\mathcal{B}$** .

Result

Let $\mathcal{B} = \{u_1, u_2, \dots, u_n\}$ and $\mathcal{C} = \{v_1, v_2, \dots, v_n\}$ be ordered bases for a vector space V and let $[I]_{\mathcal{C} \leftarrow \mathcal{B}}$ be the change of basis matrix from $\mathcal{B}$ to $\mathcal{C}$. Then

- ① $[I]_{\mathcal{C} \leftarrow \mathcal{B}}[x]_{\mathcal{B}} = [x]_{\mathcal{C}}$ for all $x \in V$;
- ② $[I]_{\mathcal{C} \leftarrow \mathcal{B}}$ is the **unique matrix** with the property that $[I]_{\mathcal{C} \leftarrow \mathcal{B}}[x]_{\mathcal{B}} = [x]_{\mathcal{C}}$ for all $x \in V$;
- ③ $[I]_{\mathcal{C} \leftarrow \mathcal{B}}$ is invertible and $([I]_{\mathcal{C} \leftarrow \mathcal{B}})^{-1} = [I]_{\mathcal{B} \leftarrow \mathcal{C}}$.

Result(Change-of-basis formula)

Let $T : V \rightarrow V$ be a linear transformation and let $\mathcal{B} = \{u_1, u_2, \dots, u_n\}$ and $\mathcal{C} = \{v_1, v_2, \dots, v_n\}$ be two ordered bases for V . Then

$$[T]_{\mathcal{B}} = [I]_{\mathcal{B} \leftarrow \mathcal{C}}[T]_{\mathcal{C}}[I]_{\mathcal{C} \leftarrow \mathcal{B}} \quad \text{or} \quad [T]_{\mathcal{B}} = ([I]_{\mathcal{C} \leftarrow \mathcal{B}})^{-1}[T]_{\mathcal{C}}[I]_{\mathcal{C} \leftarrow \mathcal{B}}$$

Example

Consider the vector space $\mathbb{R}^2$ over $\mathbb{R}$ and its two ordered bases $\mathcal{B} = \left\{ \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \end{bmatrix} \right\}$ and $\mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$. Find $[I]_{\mathcal{C} \leftarrow \mathcal{B}}$ and $[I]_{\mathcal{B} \leftarrow \mathcal{C}}$ and hence verify $([I]_{\mathcal{C} \leftarrow \mathcal{B}})^{-1} = [I]_{\mathcal{B} \leftarrow \mathcal{C}}$.

Solution

Because $I \left(\begin{bmatrix} 4 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $I \left(\begin{bmatrix} 5 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} 5 \\ 3 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, we have

$$[I]_{C \leftarrow B} = \begin{bmatrix} 4 & 5 \\ 2 & 3 \end{bmatrix}.$$

Since $I \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{3}{2} \begin{bmatrix} 4 \\ 2 \end{bmatrix} - \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ and $I \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -\frac{5}{2} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, we get

$$[I]_{B \leftarrow C} = \begin{bmatrix} \frac{3}{2} & -\frac{5}{2} \\ -1 & 2 \end{bmatrix}.$$

Also

$$[I]_{C \leftarrow B} [I]_{B \leftarrow C} = \begin{bmatrix} 4 & 5 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} \frac{3}{2} & -\frac{5}{2} \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Example

Find the change of basis matrix $[I]_{\mathcal{C} \leftarrow \mathcal{B}}$ and $[I]_{\mathcal{B} \leftarrow \mathcal{C}}$ for the ordered bases $\mathcal{B} = \{1, x, x^2\}$ and $\mathcal{C} = \{1+x, x+x^2, 1+x^2\}$ of $\mathbb{R}_2[x]$. Then find the coordinate vector of $p(x) = 1 + 2x - x^2$ with respect to the basis $\mathcal{C}$.

Solution

Because

$$I(1) = 1 = \frac{1}{2}(1+x) - \frac{1}{2}(x+x^2) + \frac{1}{2}(1+x^2)$$

$$I(x) = x = \frac{1}{2}(1+x) + \frac{1}{2}(x+x^2) - \frac{1}{2}(1+x^2)$$

$$I(x^2) = x^2 = -\frac{1}{2}(1+x) + \frac{1}{2}(x+x^2) + \frac{1}{2}(1+x^2),$$

We have $[I]_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}$. Again,

$$I(1+x) = 1+x = 1 \cdot 1 + 1 \cdot x + 0 \cdot x^2$$

$$I(x+x^2) = x+x^2 = 0 \cdot 1 + 1 \cdot x + 1 \cdot x^2$$

$$I(1+x^2) = 1+x^2 = 1 \cdot 1 + 0 \cdot x + 1 \cdot x^2$$

We get $[I]_{\mathcal{B} \leftarrow \mathcal{C}} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. Given $p(x) = 1 + 2x - x^2$ and therefore

$$[p(x)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}.$$

So, coordinate vector of $p(x)$ with respect to basis $\mathcal{C}$

$$[p(x)]_{\mathcal{C}} = [I]_{\mathcal{C} \leftarrow \mathcal{B}}[p(x)]_{\mathcal{B}} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

Exercise 37:

Let $\beta = \{v_1, v_2, v_3\}$ be a basis for $\mathbb{R}^3$ consisting of $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Let T be the linear transformation on $\mathbb{R}^3$ given by the matrix

$$[T]_{\beta} = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 3 \\ -1 & 1 & 1 \end{bmatrix}$$

Let $\alpha = \{e_1, e_2, e_3\}$ be the standard basis of $\mathbb{R}^3$. Find the transition matrix $[I]_{\beta \leftarrow \alpha}$ and $[T]_{\alpha}$.

Ans: $[I]_{\beta \leftarrow \alpha} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$ and $[T]_{\alpha} = \frac{1}{2} \begin{bmatrix} 4 & 2 & 2 \\ 3 & -1 & 1 \\ -1 & 1 & 7 \end{bmatrix}$.

Similarity

Similar Matrices

Let A and B be two square matrices of size $n \times n$ then A is said to be **Similar** to B if there exists a non-singular matrix P of size $n \times n$ such that $B = P^{-1}AP$.

Remark

Note that if A is similar to B , then B is also similar to A . Thus we simply say that A and B are similar matrices.

Example

Let $A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ then

$$B = P^{-1}AP$$

where $P = \begin{bmatrix} 1 & -1 \\ -3 & 1 \end{bmatrix}$ and hence A and B are similar matrices.

Properties of Similar Matrices

Following are some important properties of similar matrices A and B :

- ✿ Two similar matrices have the same ranks but converse need not be true.

Counter example: Consider $A = I_2$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Then A and B have the same rank 2 but they are not similar.

- ✿ Two similar matrices have same characteristic polynomials but converse need not be true.

Counter example: Let $A = I_2$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. Then both A and B have the same characteristic polynomial $(\lambda - 1)^2$ but they are not similar.

- ✿ Two similar matrices have the same eigenvalues but converse need not be true.

Counter example: Let $A = I_2$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. Then both A and B have the same eigenvalue 1 but they are not similar.

Properties of Similar Matrices cont. . .

- ✿ Two similar matrices have same determinant but converse need not be true.

Counter example: Let $A = I_2$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$. Then both A and B have the same determinant 1 but they are not similar.

- ✿ Two similar matrices have same trace but converse need not be true.

Counter example: Let $A = I_2$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$. Then both A and B have same trace 2 but they are not similar.

- ✿ If A and B are two similar matrices, then A^m and B^m are also similar matrices for any positive integer m .

- ✿ A matrix and its transpose matrix are similar, i.e., A and A^T are similar.

Result

Matrices associated with same linear transformation are similar.

Exercise 38:

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by

$$T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} x + 2y + z \\ -y \\ x + 4z \end{bmatrix}$$

Let α be the standard basis, and let $\beta = \{v_1, v_2, v_3\}$ be another ordered basis

consisting of $v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ for $\mathbb{R}^3$. Find the associated

matrix of T with respect to α and the associated matrix of T with respect to β . Are they similar? If yes, find a nonsingular matrix P such that $[T]_\beta = P^{-1}[T]_\alpha P$.

Exercise 39:

Let $D : \mathbb{R}_2[x] \rightarrow \mathbb{R}_2[x]$ be the differential operator. Find the matrix of D with respect to $E = \{1, x, x^2\}$ and with respect to $F = \{1, 2x, 4x^2 - 2\}$. By finding a Nonsingular matrix P , show that these two matrices are similar.

Exercise 40:

Let $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by $L \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} x + 3y + z \\ 2x - y + 3z \\ y - z \end{bmatrix}$. Find the matrix of L

with respect to the basis $\left\{ \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$, and show it is similar to the matrix with respect to the standard basis.

Exercise 41:

- (a) Is the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ similar to the matrix $B = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$? **No**
- (b) Is the matrix $A = \begin{bmatrix} 0 & 1 \\ 5 & 3 \end{bmatrix}$ similar to the matrix $B = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$? **No**
- (c) Is the matrix $A = \begin{bmatrix} -1 & 6 \\ -2 & 6 \end{bmatrix}$ similar to the matrix $B = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$? **Yes**
- (d) Is the matrix $A = \begin{bmatrix} -1 & 6 \\ -2 & 6 \end{bmatrix}$ similar to the matrix $B = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$? **Yes**

Application of Linear transformations in Computer graphics

- ❁ Computer graphics essentially consist of a large number of basic geometric shapes such as triangles or quadrilaterals, defined by their vertices (corners).
- ❁ A picture or an image on a screen usually consists of a number of points, lines or curves connecting some of them, and information about how to fill the regions bounded by the lines and curves. Assuming that the computer has information about how to connect the points and curves, a figure can be defined by a list of points.
- ❁ A computer graphics can be represented by a matrix with coordinates of the vertices.
- ❁ The ability to manipulate these shapes efficiently is a fundamental necessity in the field of computer graphics.
- ❁ Linear transformations can be used to generate complex shapes from basic shapes through effects such as shearing and reflection, and can also be used to rotate and translate shapes in order to create a sequence of images needed for an animated scene.

Geometric Transformations

✿ Reflection Through the x-axis:

From figure, it is clear that

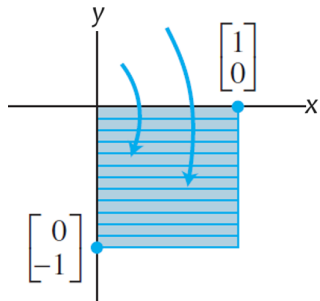
$$R_x \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

$$R_x \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, the matrix of reflection through x-axis is

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



❁ Reflection Through the y-axis:

From figure, it is clear that

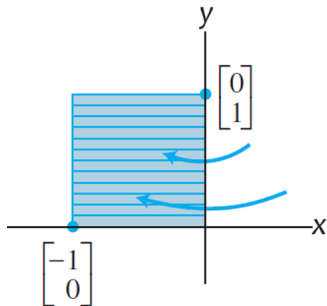
$$R_y \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix} = (-1) \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

$$R_x \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, the matrix of reflection through x-axis is

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$



❁ Reflection Through the line $y = x$:

From figure, it is clear that

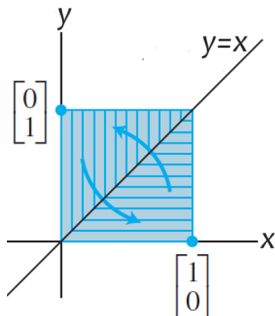
$$R_{y=x} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

$$R_{y=x} \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, the matrix of reflection through x-axis is

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



❁ Reflection Through the line $y = x$:

From figure, it is clear that

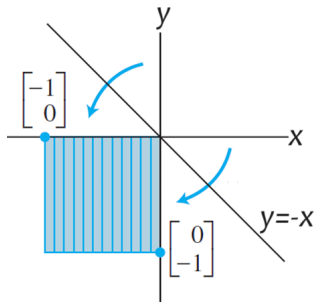
$$R_{y=x} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

$$R_{y=-x} \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix} = (-1) \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, the matrix of reflection through x-axis is

$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$



❁ Reflection Through the origin:

From figure, it is clear that

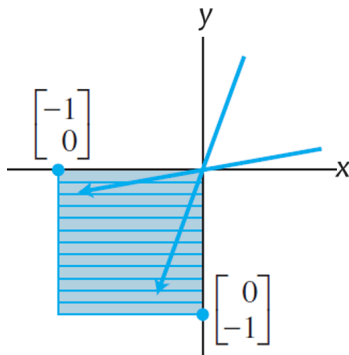
$$R_o \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix} = (-1) \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

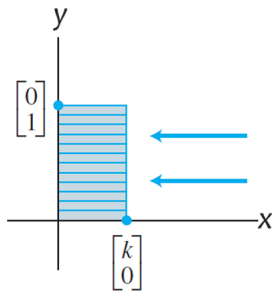
$$R_o \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, the matrix of reflection through x-axis is

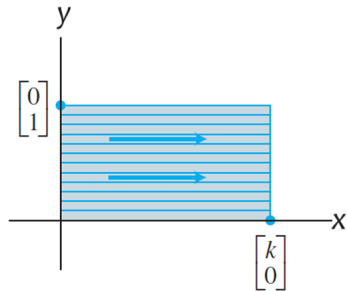
$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$



✿ Horizontal contraction and expansion:



$$0 < k < 1$$



$$k > 1$$

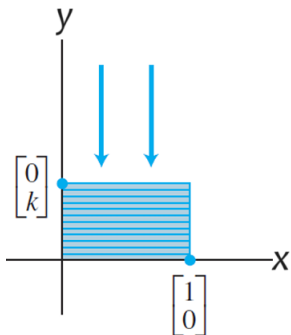
From figure, it is clear that

$$H_c \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} k \\ 0 \end{bmatrix} = k \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad H_c \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

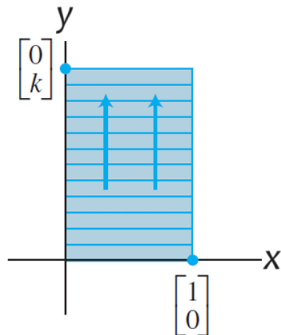
Therefore, the matrix of Horizontal contraction and expansion

$$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$$

✿ Vertical contraction and expansion:



$$0 < k < 1$$



$$k > 1$$

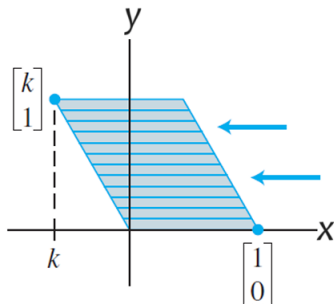
From figure, it is clear that

$$V_c \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad V_c \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ k \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + k \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

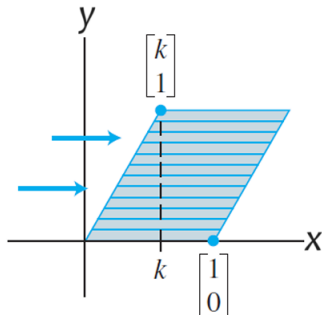
Therefore, the matrix of Vertical contraction and expansion

$$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$$

✿ Horizontal shear:



$k < 0$



$k > 0$

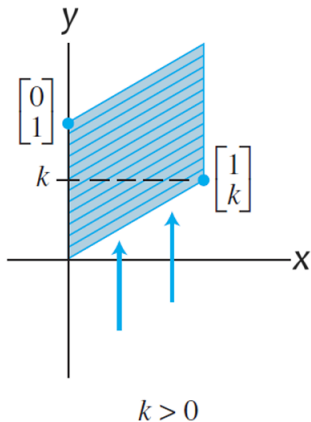
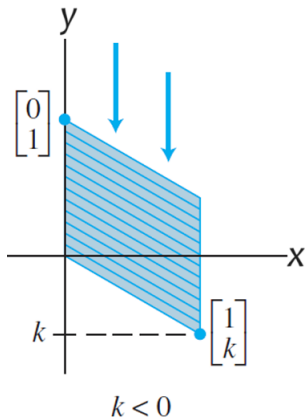
From figure, it is clear that

$$H_s \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad H_s \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} k \\ 1 \end{bmatrix} = k \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, the matrix of Horizontal shear

$$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

✿ Vertical shear:



From figure, it is clear that

$$V_s \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ k \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + k \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad V_s \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, the matrix of Vertical shear

$$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

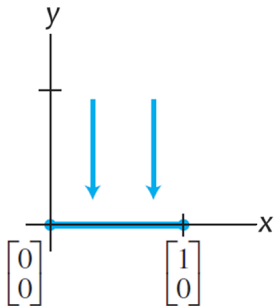
❁ Projection onto the x-axis:

From figure, it is clear that

$$P_x \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

$$P_x \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Therefore, the matrix of Projection onto the x-axis is

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

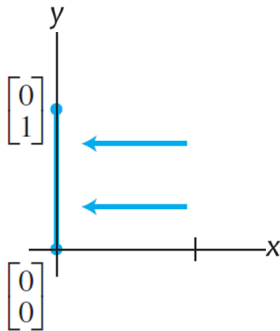
❁ Projection onto the y-axis:

From figure, it is clear that

$$P_y \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

$$P_y \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Therefore, the matrix of Projection onto the y-axis is

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

✿ Rotation by an angle ϕ :

From figure, it is clear that

$$\begin{aligned}R_{\phi} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) &= \begin{bmatrix} \cos \phi \\ \sin \phi \end{bmatrix} \\&= \cos \phi \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \sin \phi \begin{bmatrix} 0 \\ 1 \end{bmatrix}\end{aligned}$$

and

$$\begin{aligned}R_{\phi} \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) &= \begin{bmatrix} \cos \left(\frac{\pi}{2} + \phi \right) \\ \sin \left(\frac{\pi}{2} + \phi \right) \end{bmatrix} = \begin{bmatrix} -\sin \phi \\ \cos \phi \end{bmatrix} \\&= -\sin \phi \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \cos \phi \begin{bmatrix} 0 \\ 1 \end{bmatrix}\end{aligned}$$

Therefore, the matrix of Rotation by an angle ϕ is

$$\begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix}$$

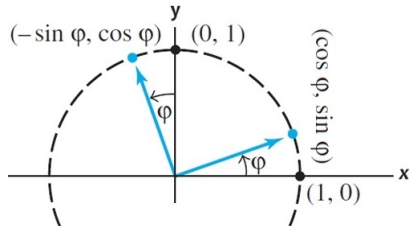
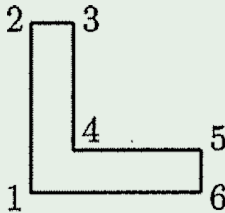


Figure 2: Rotation by an angle ϕ .

Example

Consider the capital letter "L" below:



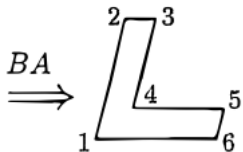
The coordinates of six vertices forms a matrix:

$$\begin{array}{l} \text{Vertices} \\ x - \text{coordinate} \\ y - \text{coordinate} \end{array} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \\ \left[\begin{array}{cccccc} 0 & 0 & 0.5 & 0.5 & 2.0 & 2.0 \\ 0 & 2.0 & 2.0 & 0.5 & 0.5 & 0 \end{array} \right] & = & A \end{array}$$

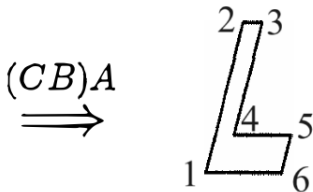
cont...

If we multiply a matrix to A , the vertices are transformed to the other set of vertices, and the line segments connecting the vertices are preserved. The horizontal shear matrix $B = \begin{bmatrix} 1 & 0.25 \\ 0 & 1 \end{bmatrix}$ transforms the matrix A to the following form, which represents new coordinates of the vertices:

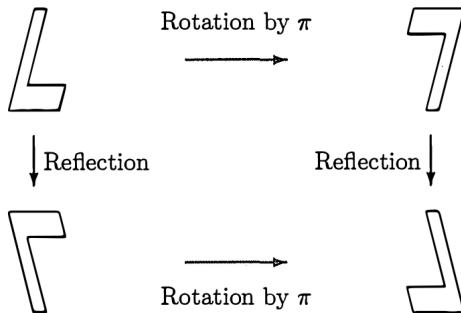
$$BA = \begin{bmatrix} 0 & 0.5 & 1.0 & 0.625 & 2.125 & \\ 0 & 2.0 & 2.0 & 0.5 & 0.5 & 0 \end{bmatrix}$$



The multiplication of the contraction matrix $C = \begin{bmatrix} 0.5 & 0 \\ 0 & 1 \end{bmatrix}$ to BA shrinks the width of BA by half.



cont...



Exercise 42:

Multiply reflections, rotations, shear, Projection and contraction or expansion to A to see how the shape of the figure changes.